Literature Review for Master Thesis

1 Introduction for Thesis

Avalanches are rapid masses of snow that move down mountain slopes due to gravity (Ancey, 2001). Snow avalanches can either be loose snow, which starts from a point in a cohesionless surface, or snow slab avalanches (Schweizer et al., 2003). In this Master Thesis, the focus will be on dry snow slab avalanches, so mostly coherent masses of snow which contain no liquid water. Snow slab avalanches consist of a cohesive slab of snow (Schweizer et al., 2003) that is released because of loading at the surface, which triggers crack propagation in a weak layer (Gaume et al., 2017). Weak layers are characterized by the grain shape of snow, which leads to lower cohesion of those layers (Föhn et al., 1998).

Snow avalanches are a major natural hazard in mountainous regions; they endanger human life as well as infrastructure (Schweizer et al., 2003). The Swiss Institute for Snow and Avalanche Research has reported an average of 22 fatalities caused by avalanches every year, with the number of non-fatal accidents being tenfold higher (SLF, 2025). These data show that it is crucial to understand how avalanches form and propagate in order to protect people and infrastructure better.

In this master's thesis, the aim is to detect avalanche slab breakoff and crack propagation using distributed acoustic sensing (DAS) cables.

In DAS, an optical signal is sent, the backscattering of this signal is measured (Shatalin et al., 1998). In distributed acoustic sensing, all points along the fibre are measured at the same time (Parker et al., 2014). The dynamic strain is estimated by measuring the phase variations of the backscattered light (Hartog, 2017). From the strain, the deformation can be obtained (Trabattoni et al., 2023). In other words, this means that a perturbation close to the DAS cable such as the breakoff of a slab avalanche causes the phase in the backscattered light to change, which is translated to deformation of the cable. mention also the cable location here, length etc

As mentioned, avalanches are a major threat to human life in mountainous regions, which is why detecting the breakoff of the snow slab is important for hazard mitigation. Detecting the breakoff of the slab means that warnings can be issued in lower parts of the mountain that an avalanche is possible. Furthermore, detecting slab breakoff with DAS is also important to further understand crack dynamics in snow as well as the factors influencing crack propagation.

A pilot early-warning system using distributed acoustic sensing has been implemented in Norway, where avalanches are detected by their seismic signature (Turquet et al., 2024), meaning when the avalanche is already moving downhill. The detectability of avalanches by DAS has also been studied in the Swiss Alps by Paitz et al., 2023 and Edme et al., 2023, but the detectability of the slab breakoff itself has not been studied.

A multitude of models are used to study different processes in snow. The SNOWPACK and CROCUS models used in the Swiss and French alps model the structure and evolution of snow (Bartelt & Lehning, 2002; Brun et al., 1992). These models were developed to understand how snow evolves under different conditions, for example a changing temperature gradient and water content and what consequences this has for the formation of weak layers where failure likely occurs and avalanches are triggered (Bartelt & Lehning, 2002; Brun et al., 1992). SNOWPACK and CROCUS are comprehensive models that take into account multiple governing equations and processes and what consequences they have for the structure of snow. While it is important to understand the influences on for example temperature gradient on the snow structure and weak layer formation and stability as a consequence, this is not the focus of the thesis. This is because these models are too complex and comprehensive for the purpose of trying to detect slab breakoff in DAS data as they model the entire snow structure. Using SNOWPACK or CROCUS to investigate how different snowpack compositions and factors influence the DAS signal is beyond the scope of this thesis.

To model crack propagation in snow, multiple numerical modeling methods have been used. Cundall and

Strack, 1979 developed the distinct element method (DEM) for granular media. This model discretizes the domain as spheres and disks, which interact with each other based on Newton's second law (Cundall & Strack, 1979). The interactions of particles are modeled by the governing equations and the resulting forces and displacements (Cundall & Strack, 1979). This method has been used by Bobillier et al., 2020, 2024; Mulak and Gaume, 2019 and many others to model crack propagation and its influencing factors. The main limitation of DEM is that the particle contacts and the corresponding equations need to be recalculated every time a grain contact is made or broken (Cundall & Strack, 1979). This means that for large, slope-scale simulations, this method becomes increasingly computationally expensive. In addition to the high computation cost, DEM also cannot account for anticrack propagation, the closure of cracks in this propagation mode is accounted for by manually removing elements from the mesh, which is not very efficient (Gaume et al., 2019).

This problem is solved by using the material point method (MPM). In this method, the material is discretized in a set of Lagrangian points, which are used to track mass, deformation and momentum (Bardenhagen et al., 2000). Because the points are not fixed, an Eulerian background mesh is needed to calculate spatial derivatives (Bardenhagen et al., 2000). This method is not only more computationally efficient on the slope-scale, it can also account for anticrack behavior and strain softening of snow (Gaume et al., 2019). This is why the material point method will also be used in the thesis to investigate slab breakoff in DAS data on a large scale. not sure if this next part is necessaryThe material point method has been used by Bergfeld et al., 2025, who modeled slope-scale experiments to understand the transition of crack propagation modes, Trottet et al., 2022 who modeled propagation saw tests to observe transitions in crack propagation speeds, as well as Gaume et al., 2019 and Gaume et al., 2018.

does this need to be more detailed? like explain what spectral elements is as well and more how salvus works? In order to understand how crack propagation and slab breakoff would look in DAS data, numerical modeling is done. This is done in SALVUS, which is a package that uses spectral elements to model seismic wavefields (Afanasiev et al., 2017).

While it is important to understand the different modeling methods and their limitations, it is also important to understand the different types of crack propagation and what factors influence them. Understanding how fast a crack can propagate and which factors influence this is important for warning, as it relates the time of the initial crack to the time where the slab breaks off depending on snow and terrain parameters.

Two main types of crack propagation are believed to be important for weak layer failure: anticrack propagation and shear crack propagation (Bergfeld et al., 2025). There are three ways in which cracks can open up: mode I, where space is opened between the crack faces under tensile loading, and modes II and III, which represent parallel shearing and diagonal shearing (Heierli et al., 2003). In anticrack propagation, the weak layer consolidates under loading (Knight & Knight, 1972), while cracks propagate in a combination of modes I-III (Heierli et al., 2003). Gaume et al., 2018 modeled anticrack propagation induced by a propagation saw test (PST) using MPM, which showed that the scale of the simulations influences the propagation direction of the cracks. For smaller scale simulations, cracks propagate from top to bottom of a slope, whereas for slope-scale simulations, cracks propagate bottom to top, which means that remote triggering of avalanches is possible (Gaume et al., 2018). This is an important consideration for the thesis because it influences where the slab breakoff would be observed in terms of DAS cable location. Other models also show that the crack propagation speed of anticracks is influenced by slope angle and curvature: concave slope areas have higher propagation speeds than convex areas of the slope (Gaume et al., 2019). Furthermore, there is a directionality in propagation speed, simulations by Gaume et al., 2019 show that cracks propagate twice as fast upslope than in the downslope direction, which is especially important when thinking about remote triggering. The slab breaks off when the slope angle exceeds the friction angle of the material (Gaume et al., 2019). Slope angle also influences which mode of crack propagation is more dominant (Rosendahl & Weißgraeber, 2020). In steeper terrains, shear modes (II and III) dominate, whereas in flatter terrain, mode I dominates the crack propagation (Rosendahl & Weißgraeber, 2020). The critical force (skier force) needed for weak layer failure is also dependent on slope angle; it decreases with increasing slope angle (Rosendahl & Weißgraeber, 2020). Snow parameters also have an influence on this critical force, as shown by simulations by Rosendahl and Weißgraeber, 2020: the thicker the slab or the higher the slab tensile strength, the better the force by the skier can be distributed on the weak layer and the more force is needed for failure.

Multiple experiments and numerical models have shown that cracks can propagate faster than the elastic wave speed in snow, so-called supershear crack propagation (Bergfeld et al., 2025; Bobillier et al., 2024; Trottet et al., 2022). Understanding what causes the transition from sub-Rayleigh to supershear crack propagation is of high importance for the thesis, because it can give an estimate of whether supershear crack propagation will occur under certain conditions, which leads to faster propagation of cracks and less time between loading and slab breakoff. In MPM models of PST, Trottet et al., 2022 observe that the occurrence of supershear cracks is dependent on slope angle: in steeper terrains, supershear propagation was observed. This can be explained by slab bending, because in steep terrains, bending of the slab is not limited to the collapse depth of the weak layer, slab tension can increase more, which leads to faster crack propagation (Trottet et al., 2022). These observations were also made by Bobillier et al., 2024 as well as Bergfeld et al., 2025. The release area of an avalanche is also tied to propagation speed: at supershear propagation speed, there is less crack arrest and therefore larger release areas (Bergfeld et al., 2025; Trottet et al., 2022). This is crucial because larger release areas also mean a larger snow volume, which has more potential for damage.

Independent of crack propagation mode, different factors influence the speed and distance at which cracks can propagate in the weak layer before failure occurs (Gaume et al., 2015, 2018; Gaume et al., 2017). As discussed above, slope angle and stiffness are important factors. Other factors include weak layer parameters: the weaker the weak layer, the slower crack propagation; the thicker the weak layer, the lower the crack propagation distance before failure (Gaume et al., 2015). The depth of the slab also has an influence on propagation distance and speed: the thicker the slab, the larger the propagation distance and velocity (Gaume et al., 2015). How fast a slab will break off is also dependent on the critical crack length needed for failure (Gaume et al., 2017). If the snowpack is more stable, which is the case when the slab has a high elastic modulus or the weak layer is thicker, the critical crack length is higher (Gaume et al., 2017) because the snow can sustain longer cracks before it fails. A very important factor in material strength is loading rate: if loading is slow, sintering can occur (Mulak & Gaume, 2019). This means that new snow contacts can form, which can strengthen the material (Mulak & Gaume, 2019). Sintering can occur if the loading is slow, which is the case when the loading occurs due to snow accumulation, in contrast to rapid loading, which happens for example due to a skier (Mulak & Gaume, 2019).

Based on the above information, there are multiple considerations necessary in order to model DAS data of crack propagation accurately. The scale of the models is an important factor, because as mentioned, it influences the direction of crack propagation. This influences at which location along or relative to the DAS cable slab breakoff will occur, which changes the measured data. Furthermore, multiple parameters influence the time passed between initial cracking and slab breakoff, these are important to consider when interpreting data. The different modes of crack propagation are also important to consider, it is possible that a transition between sub-Rayleigh and supershear crack propagation can be observed in the data. The loading rate can also be considered, because slower loading leads to stabilizing processes such as sintering.

here i elaborate the research questions, but I don't know which one I will focus on yet

- Different DAS signatures for different propagation modes (sub-raleigh vs supershear)
- models based on loading rate to see if DAS data looks different
- Distinction of processes: Weak layer colapse, crown and staunchwall fractures
- Relation to snow properties in models
- relationship between crack propagation and release area

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